Interferenza
1/29/23 4:00 PM

Considero 2 onde piane sin nello stesso mezzo:

$$\begin{cases} \omega_1 = \omega_2 = \omega \\ \omega = \frac{2\pi}{\tau} \quad \lambda \nu = v \quad \lambda = \frac{v}{\nu} \ ! \\ K_1 = K_2 = K = \frac{2\pi}{\lambda} \end{cases}$$

$$\phi = (K \cdot x + \varphi)$$

$$\begin{cases} E_1 = E_{01} e^{i(\omega t - K \cdot x_1 - \varphi_1)} = E_{01} e^{i\omega t} e^{-i(Kx_1 + \varphi_1)} = E_{01} e^{i\omega t} e^{i\phi_1} \\ E_2 = E_{02} e^{i(\omega t - K \cdot x_2 - \varphi_2)} = E_{01} e^{i\omega t} e^{-i(Kx_2 + \varphi_2)} = E_{02} e^{i\omega t} e^{i\phi_2} \end{cases}$$

$\Rightarrow E_{tot} = \bar{E}_1 + \bar{E}_2$

NB Per 1 onda $\bar{I} = \frac{1}{2}\sqrt{\frac{\varepsilon}{\mu}}\, E_0^2$

$\Rightarrow I \propto |E_1 + E_2|^2 = E_{01}^2 + E_{02}^2 + 2E_{01}E_{02}\cos(\phi_2 - \phi_1)$

- Se $(\phi_2 - \phi_1) = cost \rightarrow$ fonti COERENTI
  $I \propto |\bar{E}_1 + \bar{E}_2|^2 \Rightarrow$ interferenza
- Se $(\phi_2 - \phi_1) \neq cost \qquad \overline{\cos(\phi_2 - \phi_1)} = 0$
  $\Rightarrow I = I_1 + I_2$

INTERFERENZA

IPOTESI
- fonti coerenti
- $\omega_1 = \omega_2 = \omega = \frac{2\pi}{\tau}$
- stessa polarizzazione

$\lambda_0 \nu = c \quad \lambda_0 = \frac{c}{\nu}$

$\lambda \nu = v = \frac{c}{n} \quad \lambda = \frac{c}{\nu n} = \frac{\lambda_0}{n}$

def: DIFFERENZA di FASE

$$\delta = \phi_2 - \phi_1 = (K_2 x_2 - K_1 x_1) = 2\pi\left(\frac{1}{\lambda_2}x_2 - \frac{1}{\lambda_1}x_1\right) = \frac{2\pi}{\lambda_0}\left(\underset{d_1}{n_2 x_2} - \underset{d_2}{n_1 x_1}\right)$$

def CAMMINO OTTICO

$d = \sum n_i x_i$ se attraverso diversi mezzi

$\xi_1 = \xi_{01} \operatorname{sen}(\omega t - K x_1) \qquad \phi_1 = K x_1 + \varphi_1$

$\xi_2 = \xi_{02} \operatorname{sen}(\omega t - K x_2) \qquad \phi_2 = K x_2 + \varphi_2$

$\xi = \xi_1 + \xi_2 = \xi_0 \operatorname{sen}(\omega t + \alpha)$

$\xi_0 = \sqrt{\xi_{01}^2 + \xi_{02}^2 + 2\xi_{01}\xi_{02}\cos\delta} \longrightarrow$

$$\begin{cases} \xi_1 = \xi_{01} \operatorname{sen}(\omega t - K x_1) = \operatorname{Im} \xi_{01} e^{i(\omega t - K x_1)} \\ \xi_2 = \xi_{02} \operatorname{sen}(\omega t - K x_2) \quad \operatorname{Im} \xi_{02} e^{i(\omega t - K x_2)} \end{cases}$$

$\xi = e^{i\omega t}\left(\xi_{01} e^{-iKx_1} + \xi_{02} e^{-iKx_2}\right)$

= Bho prendilo per vero e imparalo a memoria

$\xi_{max} = |\xi_{01} + \xi_{02}|$ se $\cos\delta = 1$

$\xi_{MIN} = |\xi_{01} - \xi_{02}|$ se $\cos\delta = -1$

$n_1 = n_2$

INTERFERENZA COSTRUTTIVA

$\delta = 2n\pi$

$\frac{2\pi}{\lambda}(x_2 - x_1) = 2n\pi \Rightarrow (x_2 - x_1) = n\lambda$

INT DISTR

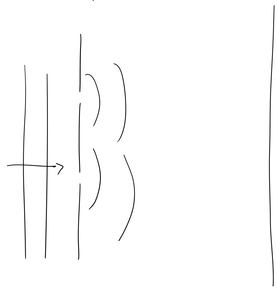
$\delta = (2n+1)\pi$

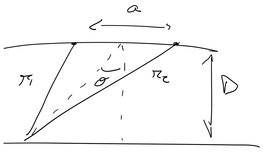
$\frac{2\pi}{\lambda}(x_2 - x_1) = (2n+1)\pi \Rightarrow x_2 - x_1 = (2n+1)\frac{\lambda}{2}$

Se $\delta = cost \quad \xi = \xi_0 \operatorname{sen}(\omega t + \alpha) \quad (\alpha = cost)$

$\Rightarrow$ no propagazione $\Rightarrow$ ONDA STAZIONARIA

ESPERIMENTO di YOUNG





A grandi distanze:

$a \ll x_1 ; x_2 \qquad x_1 \cong x_2 \qquad \xi_{01} \cong \xi_{02}$

$\xi_0 = \sqrt{\xi_{01}^2 + \xi_{02}^2 + 2\xi_{01}\xi_{02}\cos\delta}$

$= \xi_{01}\sqrt{2(1 + \cos\delta)} = 2\,\xi_{01}\cos\frac{\delta}{2}$

$\Rightarrow$

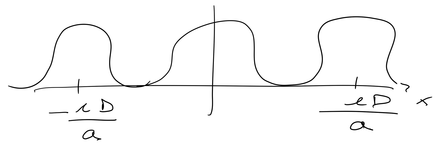
$I = I_0 \cos^2\frac{\delta}{2} \qquad I_0 = I_{max}$

$x_2 - x_1 = a \operatorname{sen}\theta \quad$ se $\theta \cong \operatorname{tg}\theta \cong \frac{x}{D}$

$\Rightarrow \delta = \frac{2\pi}{\lambda}(x_2 - x_1) = \frac{2\pi a \operatorname{sen}\theta}{\lambda} \cong \frac{2\pi a}{\lambda}\frac{x}{D}$

$I = I_{max}$ quando

$\frac{\delta}{2} = n\pi \Rightarrow \frac{\pi a x}{\lambda D} = n\pi \Rightarrow x = \frac{n\lambda D}{a}$



INTERFERENZA di N SORGENTI

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